Apart from the regular HCI infrastructure, some start-ups are also coming up with innovative solutions, which some trend-watchers describe as HCI 2.0! NetApp, for example, has an option for those who want to keep their servers and storage separate—either to share the storage with non-HCI systems or to off-load certain tasks to dedicated servers. NetApp HCI uses SolidFire technology to deliver clusters with dedicated storage and compute nodes.

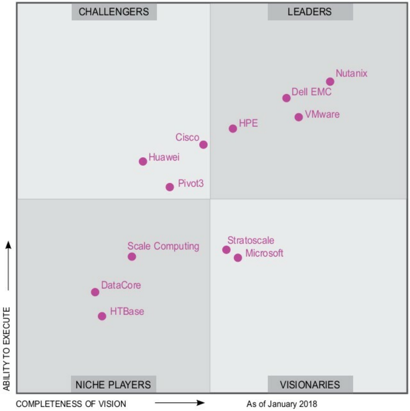
Companies like Pivot3, Atlantis Computing and Maxta also offer pure software HCI solutions.

AI and machine learning may create demand

It is interesting to note that Dell EMC is pushing NVIDIA GPU accelerators in its HCI solutions not just for video processing but also for running machine learning algorithms. Chad Dunn, who leads product management and marketing for Dell EMC's HCI portfolio, explains in a media interview, "All the HCI solutions have a hypervisor and generally in HPC you're going for a bare-metal performance and you want as close to real-time operations as you possibly can. Where that could start to change is in machine learning and artificial intelligence (AI). You typically think of the Internet-of-Things intelligent edge use cases. There's so much data being generated by the IOT that the data itself is not valuable. The insight that the data provides you is exceptionally valuable, so it makes a lot of sense not to bring that data all the way back to the core. You want to put the data analytics and decision-making of that data as close to the devices as you can, and then take the valuable insight and leave the particularly worthless data out where it is. What becomes interesting is that the form factors that HCI can get to are relatively small. Where the machine learning piece comes in is what we expect to see and what we're starting to see is people looking to leverage the graphical processor units in these platforms."

Incidentally, in November 2017, Nutanix's president Sudheesh Nair also spoke about how AI and machine learning are becoming extremely important for customers, at the company's *.Next* user conference. He explained: "If we have a customer who is prototyping an autonomous car, that car generates almost 16TB of data per day, per city. An oil rig generates around 100TB per rig in the middle of an ocean with no connectivity. There is no way you can bring this data to the cloud—you have to take the cloud to the data, to where the data is being created. But moving information from the datacentre to the cloud creates manageability issues, and AI can be a better option. That's where machine learning and artificial intelligence have a big part to play."

Companies have to work out ways to store all that information such that you can easily access it and run analytical models to get predictive behaviour. This surge in demand for AI and machine learning provides a real opportunity for HCI.



The growth in software-based hyper-convergence has re-positioned many players in the Gartner Magic Quadrant this year (Source: Gartner)

Five good reasons why CIOs are moving to HCI and why IT guys should stay on top of the tech

Although hyper-convergence as a concept has been around for more than five years, it is obviously gaining a lot of momentum now with everybody talking about it and the spate of acquisitions. And it is not without real benefits as it:

1. Is 'really' quick and easy
2. Does not require a big team to manage it
3. Reduces cost of ownership
4. Makes it easy to launch and scale on-the-go
5. Improves reliability and flexibility of the data centre

As we mentioned earlier, HCI is not the panacea for all your infrastructure management problems. But if you need new infrastructure or have to replace an existing one, do ask yourself whether an HCI system can do the job. If it can, go for it because it can ease your admin and cost headaches quite a bit! EFY